Web Scraping

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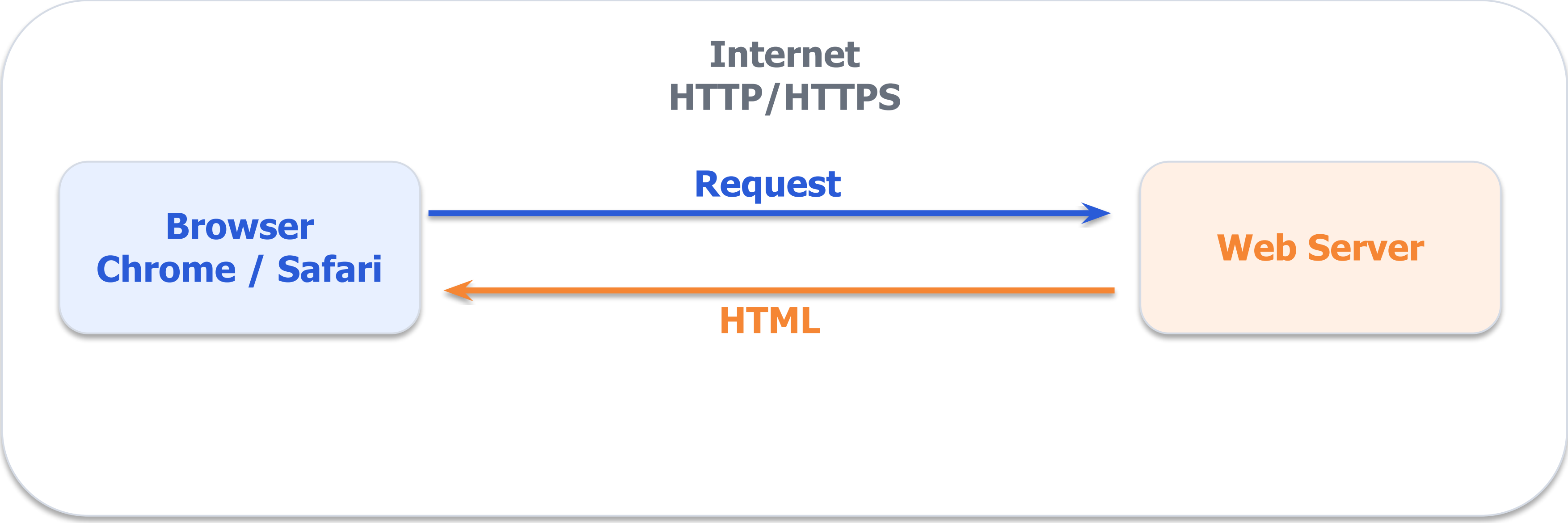
Plan for Today

1. How does the internet work?
2. How does HTML work?
3. How does Web Scraping work?
4. Social Media

How does the internet work?



1. Request
“Please give me this URL.”
Includes headers, cookies etc

2. Response
HTML, CSS, JavaScript, images
Plus status code like 200 or 404

3. Rendering
Browser turns files into the page

—

HTML

- HyperText Markup Language
- HTML uses tags to label content, common tags include:
  - <h1> for headings
  - <p> for paragraphs
  - <img> for images
  - <a> for links
- HTML works with CSS
  - CSS changes how HTML looks, also help us locate data
  - Often defined as: class="expertise"

```
<section class="member-card">
  <h5 class="member-list-name">
    何俊霆
  </h5>
  <p class="expertise">計算方法 / 政治傳播</p>
</section>
```

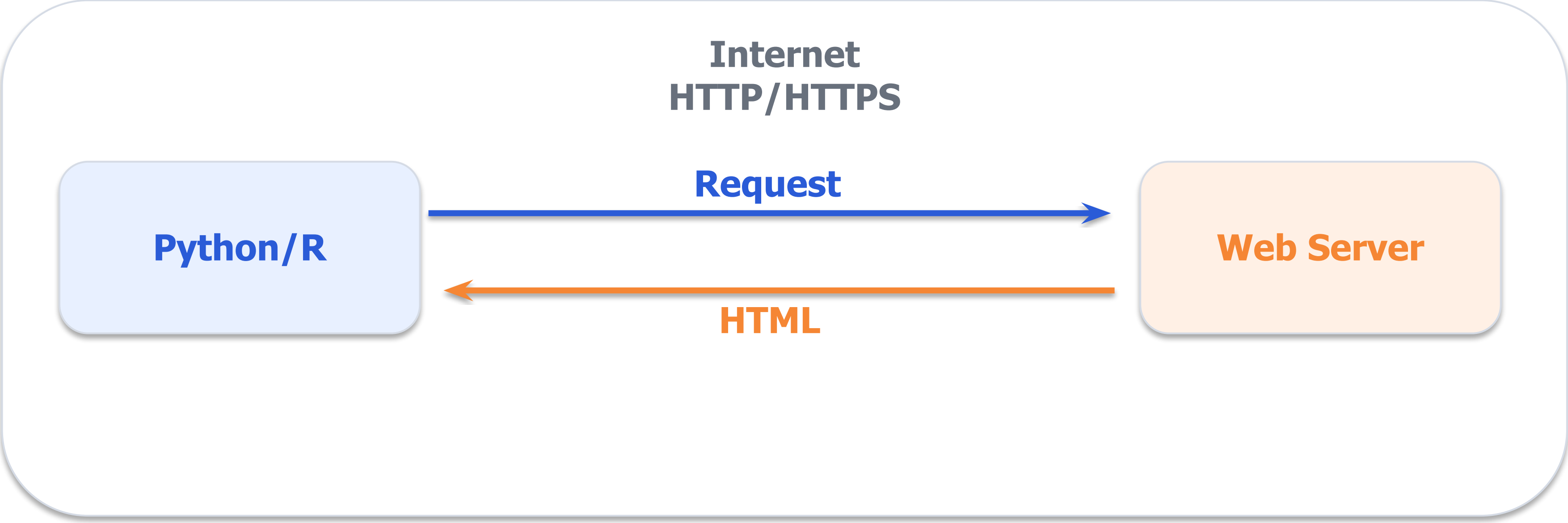
```
.member-list-name {
  font-size: 20px;
  font-weight: bold;
  color: #222;
}

.expertise {
  color: #666;
}
```

Example

https://dcat.nycu.edu.tw/members/faculty-and-staff/

How does the web scraping work?



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3. Parsing
Extract information from HTML

Web Scraping Logic

1. Identify what you want to look for (eg names)
2. Look for pattern: Do they all have the same font size/colour? Some tags? Same location in page elements?
3. In modern web pages, there is probably a CSS class for everything. Right Click -> Inspect: Look for class="something"
4. Use tools to help you identify what you want eg. SelectorGadget, ChatGPT (after this workshop you will know exactly what to ask)


Time for Python

Some Afterthoughts

- For websites that do procedural generation:

  Use Playwright to control a zombie web browser
- Most websites have RSS:

  https://service.ltn.com.tw/RSS
- If possible, use existing tools…

Social Media

- Pro-tip: Just don’t scrape it, you will get yourself blocked
- Let me show you Zeeschuimer:

  https://github.com/digitalmethodsinitiative/zeeschuimer
- Use Zeeschuimer + your browser to get all posts

  Then use Python/R to parse the output and download whatever you need
- Use https://publicdatalab.github.io/zeehaven/ to get spreadsheet